Newman-Penrose Formalism Calculations

Stephani et al., Equation (14.18b)

Metric

$$ds^2 = \left(-\frac{\left(\frac{\sinh\left(\frac{1}{2}\,t\right)}{\cosh\left(\frac{1}{2}\,t\right)}\right)^{2\,\sqrt{-M\lambda^4+1}}\sinh\left(t\right)^2}{\lambda^2}\right)\mathrm{d}t\otimes\mathrm{d}t + \left(\frac{1}{\lambda^2\left(\frac{\sinh\left(\frac{1}{2}\,t\right)}{\cosh\left(\frac{1}{2}\,t\right)}\right)^{2\,\sqrt{-M\lambda^4+1}}}\right)\mathrm{d}x\otimes\mathrm{d}x + \left(\frac{\left(\frac{\sinh\left(\frac{1}{2}\,t\right)}{\cosh\left(\frac{1}{2}\,t\right)}\right)^{2\,\sqrt{-M\lambda^4+1}}\sinh\left(t\right)^2}{\lambda^2}\right)\mathrm{d}y\otimes\mathrm{d}y + \left(\frac{\left(\frac{\sinh\left(\frac{1}{2}\,t\right)}{\cosh\left(\frac{1}{2}\,t\right)}\right)^{2\,\sqrt{-M\lambda^4+1}}\sinh\left(t\right)^2\sinh\left(y\right)^2}{\lambda^2}\right)\mathrm{d}z\otimes\mathrm{d}z$$

Coordinates

$$(x^\mu) = (t, x, y, z)$$

Null Tetrad

Covariant components

$$l = \left(\frac{\sqrt{2}\tanh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}\sinh\left(t\right)}{2\,\lambda}\right)\,dt + \left(\frac{\sqrt{2}\tanh\left(\frac{1}{2}\,t\right)^{-\sqrt{-M\lambda^4+1}}}{2\,\lambda}\right)\,dx$$

$$n = \left(\frac{\sqrt{2}\tanh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}\sinh\left(t\right)}{2\,\lambda}\right)\,dt + \left(-\frac{\sqrt{2}\tanh\left(\frac{1}{2}\,t\right)^{-\sqrt{-M\lambda^4+1}}}{2\,\lambda}\right)\,dx$$

$$m = \left(\frac{\sqrt{2}\tanh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}\sinh\left(t\right)}{2\,\lambda}\right)\,dy + \left(\frac{i\,\sqrt{2}\tanh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}\sinh\left(t\right)\sinh\left(y\right)}{2\,\lambda}\right)\,dz$$

$$\overline{m} = \left(\frac{\sqrt{2}\tanh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}\sinh\left(t\right)}{2\,\lambda}\right)\,dy + \left(-\frac{i\,\sqrt{2}\tanh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}\sinh\left(t\right)\sinh\left(y\right)}{2\,\lambda}\right)\,dz$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}\lambda\cosh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}}{2\,\sinh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}\sinh\left(t\right)}\right)\,dt + \left(\frac{\sqrt{2}\lambda\sinh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}}{2\,\cosh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}}\right)\,dx$$

$$n = \left(-\frac{\sqrt{2}\lambda\cosh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}}{2\,\sinh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}\sinh\left(t\right)}\right)\,dt + \left(-\frac{\sqrt{2}\lambda\sinh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}}{2\,\cosh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}}\right)\,dx$$

$$m = \left(\frac{\sqrt{2}\lambda\cosh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}}{2\,\sinh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}\sinh\left(t\right)}\right)\,dy + \left(\frac{i\,\sqrt{2}\lambda\cosh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}}{2\,\sinh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}\sinh\left(t\right)\sinh\left(y\right)}\right)\,dz$$

$$\overline{m} = \left(\frac{\sqrt{2}\lambda\cosh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}}{2\,\sinh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}\sinh\left(t\right)}\right)\,dy + \left(-\frac{i\,\sqrt{2}\lambda\cosh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}}{2\,\sinh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}\sinh\left(t\right)\sinh\left(y\right)}\right)\,dz$$

Directional Derivatives

$$D = -\frac{\sqrt{2}\lambda\cosh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}}{2\,\sinh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}\sinh\left(t\right)}\,\partial_t + \frac{\sqrt{2}\lambda\sinh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}}{2\,\cosh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}}\,\partial_x$$

$$\Delta = -\frac{\sqrt{2}\lambda\cosh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}}{2\,\sinh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}\sinh\left(t\right)}\,\partial_t - \frac{\sqrt{2}\lambda\sinh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}}{2\,\cosh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}}\,\partial_x$$

$$\delta = \frac{\sqrt{2}\lambda\cosh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}}{2\,\sinh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}\sinh\left(t\right)}\,\partial_y + \frac{i\,\sqrt{2}\lambda\cosh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}}{2\,\sinh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}\sinh\left(t\right)\sinh\left(y\right)}\,\partial_z$$

$$\bar{\delta} = \frac{\sqrt{2}\lambda \cosh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}}{2\,\sinh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}\sinh\left(t\right)}\,\partial_y - \frac{i\,\sqrt{2}\lambda \cosh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}}{2\,\sinh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}\sinh\left(t\right)\sinh\left(y\right)}\,\partial_z$$

Spin Coefficients

$$\rho = \frac{\sqrt{2}\lambda \cosh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}\sinh\left(\frac{1}{2}\,t\right)^{-\sqrt{-M\lambda^4+1}}\cosh\left(t\right)}{2\,\sinh\left(t\right)^2} + \frac{\sqrt{2}\sqrt{-M\lambda^4+1}\lambda \cosh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}\sinh\left(\frac{1}{2}\,t\right)^{-\sqrt{-M\lambda^4+1}}}{4\,\cosh\left(\frac{1}{2}\,t\right)\sinh\left(\frac{1}{2}\,t\right)\sinh\left(t\right)}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}\lambda \cosh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}\sinh\left(\frac{1}{2}\,t\right)^{-\sqrt{-M\lambda^4+1}}\cosh\left(t\right)}{2\,\sinh\left(t\right)^2} - \frac{\sqrt{2}\sqrt{-M\lambda^4+1}\lambda \cosh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}\sinh\left(\frac{1}{2}\,t\right)^{-\sqrt{-M\lambda^4+1}}}{4\,\cosh\left(\frac{1}{2}\,t\right)\sinh\left(\frac{1}{2}\,t\right)\sinh\left(t\right)}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\sqrt{2}\lambda \cosh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}\cosh\left(y\right)}{4\,\sinh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}\sinh\left(t\right)\sinh\left(y\right)}$$

$$\beta = \frac{\sqrt{2}\lambda \cosh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}\cosh\left(y\right)}{4\,\sinh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)}\sinh\left(t\right)\sinh\left(y\right)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = \frac{\sqrt{2}\sqrt{-M\lambda^4 + 1}\lambda \cosh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)} \sinh\left(\frac{1}{2}\,t\right)^{-\sqrt{-M\lambda^4+1}}}{8\,\cosh\left(\frac{1}{2}\,t\right)\sinh\left(\frac{1}{2}\,t\right)\sinh\left(t\right)}$$

$$\gamma = -\frac{\sqrt{2}\sqrt{-M\lambda^4 + 1}\lambda \cosh\left(\frac{1}{2}\,t\right)^{\left(\sqrt{-M\lambda^4+1}\right)} \sinh\left(\frac{1}{2}\,t\right)^{-\sqrt{-M\lambda^4+1}}}{8\,\cosh\left(\frac{1}{2}\,t\right)\sinh\left(\frac{1}{2}\,t\right)\sinh\left(t\right)}$$

Ricci Tensor Components

$$\begin{aligned}
\Phi_{00} = {} & \frac{M\lambda^6 \cosh\left(\frac{1}{2}\,t\right)^{2\,\sqrt{-M\lambda^4+1}}}{8\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\,\sqrt{-M\lambda^4+1}}\sinh\left(t\right)^2} - \frac{\sqrt{-M\lambda^4+1}\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\,\sqrt{-M\lambda^4+1}}\cosh\left(\frac{1}{2}\,t\right)\cosh\left(t\right)\sinh\left(\frac{1}{2}\,t\right)}{4\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\,\sqrt{-M\lambda^4+1}}\sinh\left(t\right)^3} \\
& + \frac{\sqrt{-M\lambda^4+1}\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\,\sqrt{-M\lambda^4+1}}\cosh\left(\frac{1}{2}\,t\right)^2}{4\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\,\sqrt{-M\lambda^4+1}}\sinh\left(t\right)^2} \\
& + \frac{\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\,\sqrt{-M\lambda^4+1}}\cosh\left(\frac{1}{2}\,t\right)^4}{2\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\,\sqrt{-M\lambda^4+1}}\sinh\left(t\right)^4} - \frac{\sqrt{-M\lambda^4+1}\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\,\sqrt{-M\lambda^4+1}}}{8\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\,\sqrt{-M\lambda^4+1}}\sinh\left(t\right)^2} \\
& - \frac{\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\,\sqrt{-M\lambda^4+1}}\cosh\left(\frac{1}{2}\,t\right)^2}{2\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\,\sqrt{-M\lambda^4+1}}\sinh\left(t\right)^4} - \frac{\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\,\sqrt{-M\lambda^4+1}}}{8\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\,\sqrt{-M\lambda^4+1}}\sinh\left(t\right)^2}
\end{aligned}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$
\begin{aligned}
\Phi_{11} = {} & \frac{M\lambda^6 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}}{16\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^2} + \frac{\sqrt{-M\lambda^4+1}\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\cosh\left(\frac{1}{2}\,t\right)\cosh(t)\sinh\left(\frac{1}{2}\,t\right)}{8\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^3} \\
& - \frac{\sqrt{-M\lambda^4+1}\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\cosh\left(\frac{1}{2}\,t\right)^2}{8\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^2} + \frac{\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\cosh\left(\frac{1}{2}\,t\right)^4}{4\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^4} \\
& + \frac{\sqrt{-M\lambda^4+1}\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}}{16\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^2} \\
& - \frac{\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\cosh\left(\frac{1}{2}\,t\right)^2}{4\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^4} - \frac{\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}}{16\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^2}
\end{aligned}
$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$
\begin{aligned}
\Phi_{22} = {} & \frac{M\lambda^6 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}}{8\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^2} - \frac{\sqrt{-M\lambda^4+1}\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\cosh\left(\frac{1}{2}\,t\right)\cosh(t)\sinh\left(\frac{1}{2}\,t\right)}{4\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^3} \\
& + \frac{\sqrt{-M\lambda^4+1}\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\cosh\left(\frac{1}{2}\,t\right)^2}{4\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^2} \\
& + \frac{\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\cosh\left(\frac{1}{2}\,t\right)^4}{2\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^4} - \frac{\sqrt{-M\lambda^4+1}\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}}{8\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^2} \\
& - \frac{\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\cosh\left(\frac{1}{2}\,t\right)^2}{2\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^4} - \frac{\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}}{8\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^2}
\end{aligned}
$$

$$\Lambda = -\frac{M\lambda^6 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}}{48\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^2} + \frac{\sqrt{-M\lambda^4+1}\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\cosh\left(\frac{1}{2}\,t\right)\cosh(t)\sinh\left(\frac{1}{2}\,t\right)}{24\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^3}$$
$$- \frac{\sqrt{-M\lambda^4+1}\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\cosh\left(\frac{1}{2}\,t\right)^2}{24\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^2} - \frac{\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\cosh\left(\frac{1}{2}\,t\right)^4}{12\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^4}$$
$$+ \frac{\sqrt{-M\lambda^4+1}\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}}{48\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^2} + \frac{\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\cosh\left(\frac{1}{2}\,t\right)^2}{12\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^4}$$
$$+ \frac{\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}}{48\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^2}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{M\lambda^6 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}}{6\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^2} - \frac{\sqrt{-M\lambda^4+1}\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\cosh\left(\frac{1}{2}\,t\right)\cosh(t)\sinh\left(\frac{1}{2}\,t\right)}{3\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^3}$$
$$- \frac{\sqrt{-M\lambda^4+1}\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\cosh\left(\frac{1}{2}\,t\right)^2}{6\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^2} - \frac{\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\cosh\left(\frac{1}{2}\,t\right)^4}{3\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^4}$$
$$+ \frac{\sqrt{-M\lambda^4+1}\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}}{12\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^2} + \frac{\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\cosh\left(\frac{1}{2}\,t\right)^2}{3\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^4}$$
$$- \frac{\lambda^2 \cosh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}}{6\left(\sinh\left(\frac{1}{2}\,t\right)^4 + \sinh\left(\frac{1}{2}\,t\right)^2\right)\sinh\left(\frac{1}{2}\,t\right)^{2\sqrt{-M\lambda^4+1}}\sinh(t)^2}$$

$$\Psi_3 = 0$$

$\Psi_4 = 0$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"